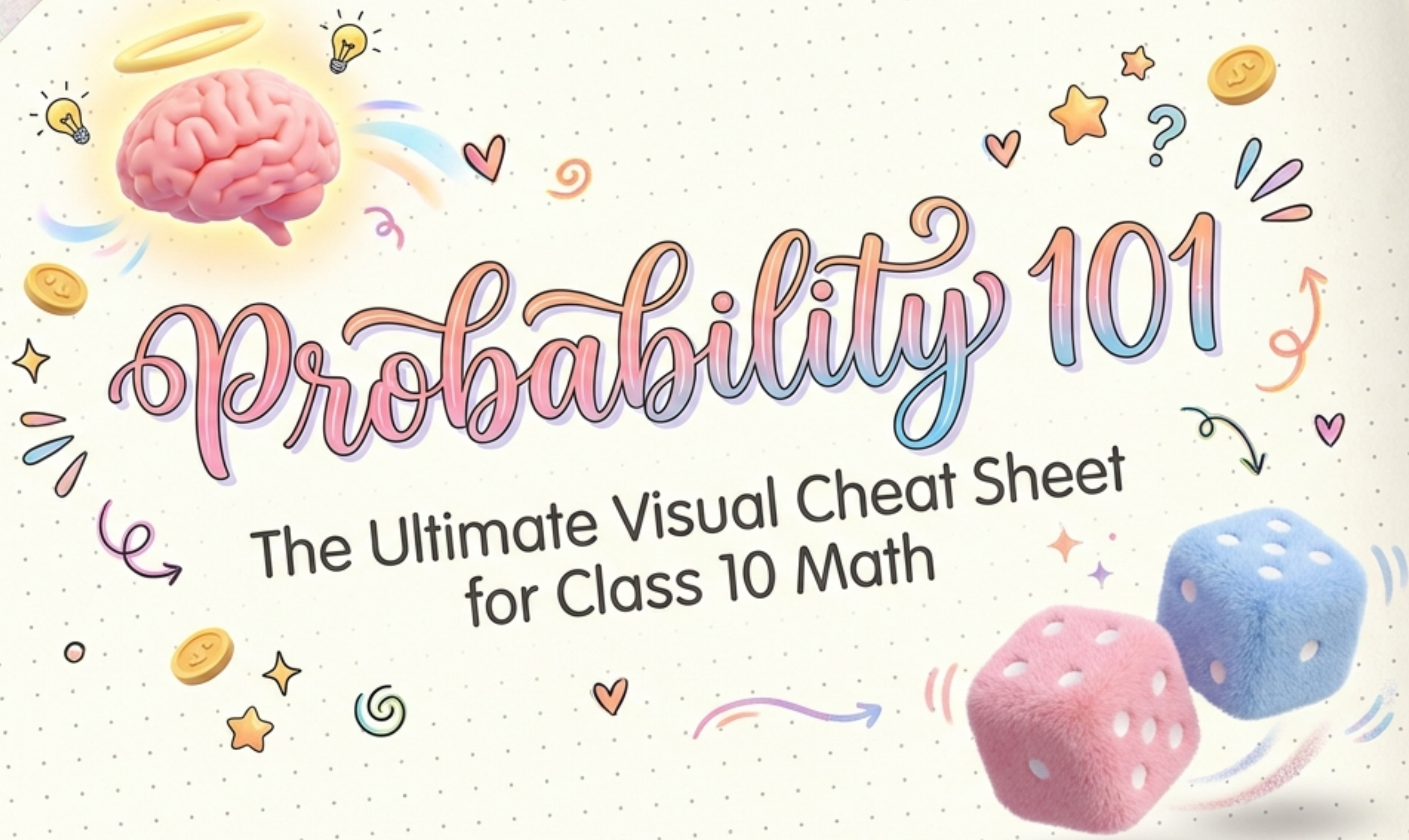




The Master Formula

$$P(E) = \frac{\text{Number of outcomes favourable to E}}{\text{Number of all possible outcomes}}$$

- Pierre Simon Laplace (1795)

Rule: Assumes all outcomes are Equally Likely!



$$0 \leq P(E) \leq 1$$

Impossible
Event



Equally
Likely
(50/50)



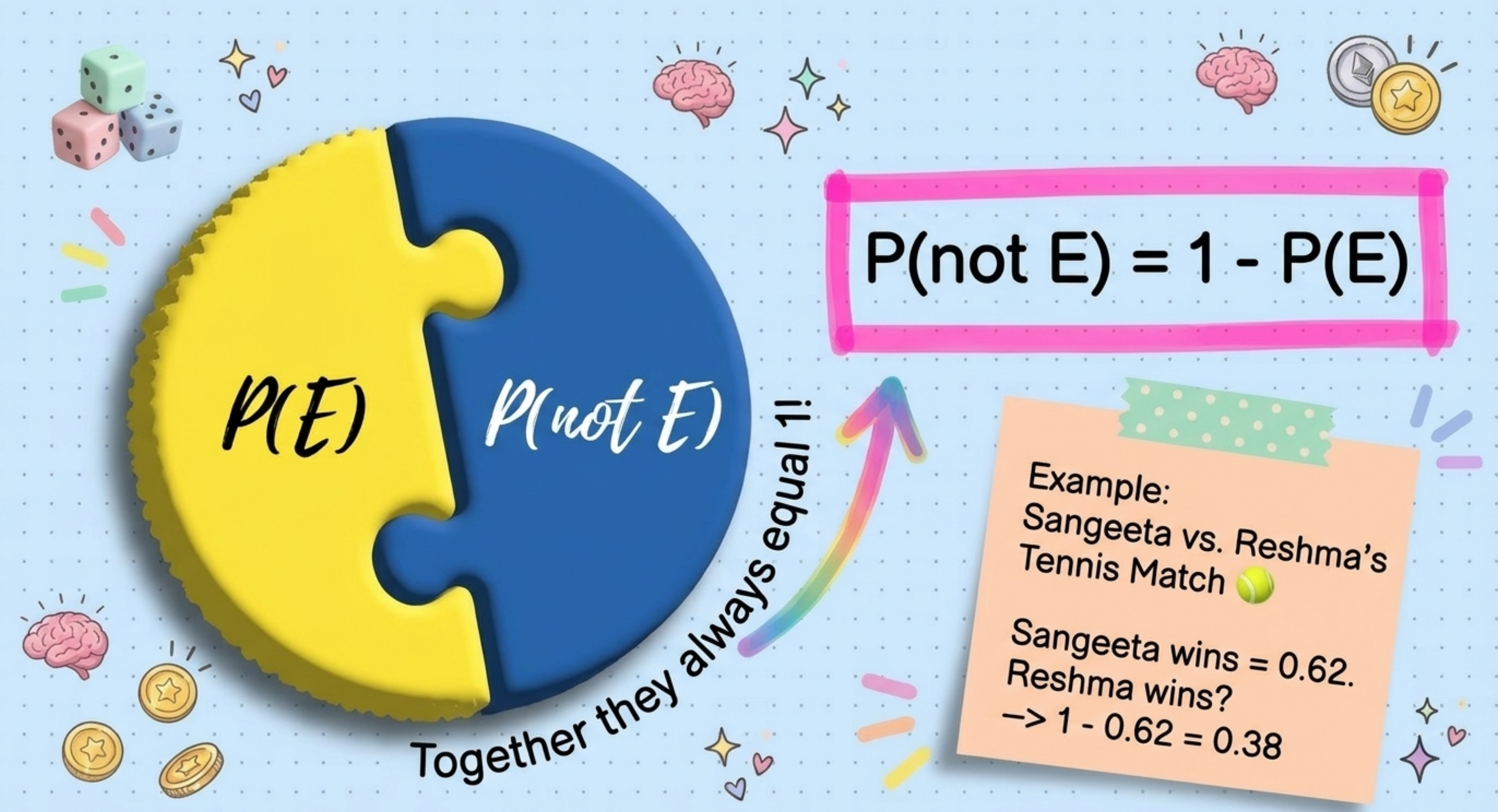
Sure /
Certain
Event



0

0.5

1



$P(E)$

$P(\text{not } E)$

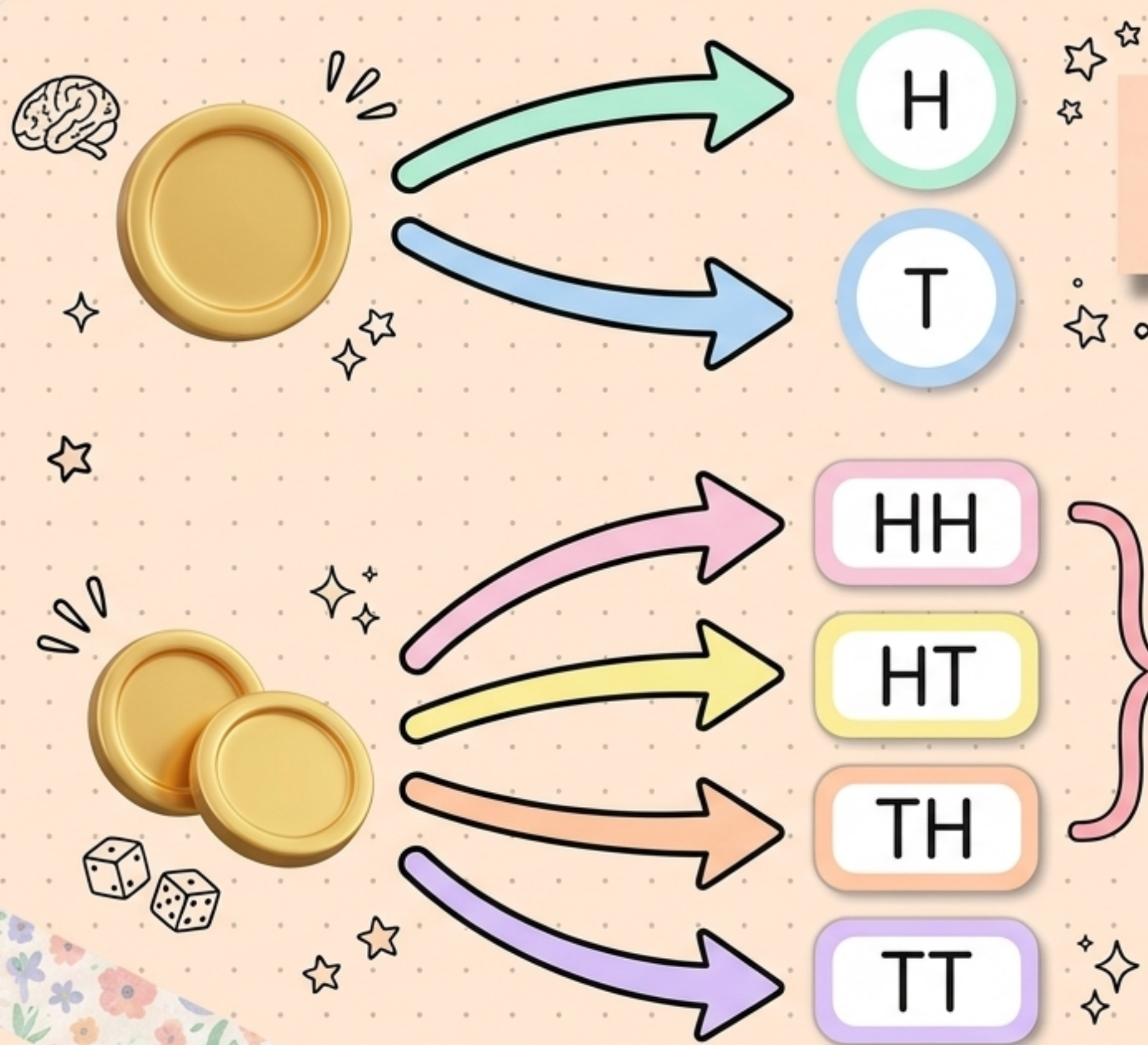
Together they always equal 1!

$$P(\text{not } E) = 1 - P(E)$$

Example:
Sangeeta vs. Reshma's
Tennis Match 🎾

Sangeeta wins = 0.62.
Reshma wins?
 $\rightarrow 1 - 0.62 = 0.38$

Sample Spaces: Coin Toss Mind Map



Sum of elementary events
 $= 1/2 + 1/2 = 1$

Q: Probability of at least
one head?

A: 3 out of 4 outcomes! $\rightarrow 3/4$

52 Cards

26 Red

26 Black

13 Hearts 

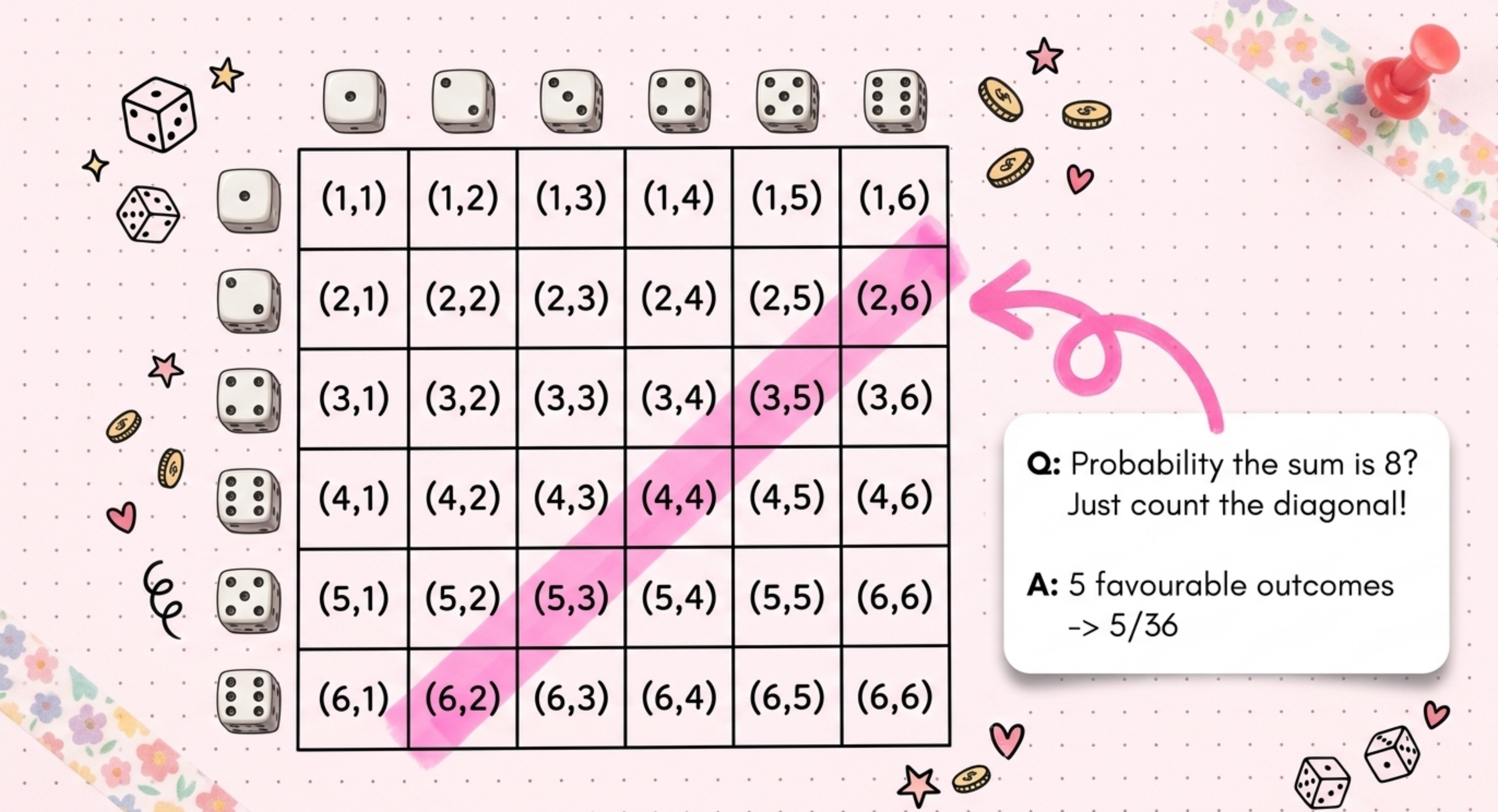
13 Diamonds 

13 Clubs 

13 Spades 



 **The Face Cards:**
Kings, Queens, and Jacks.
There are exactly 3 per suit,
making 12 total in the deck!



						
	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(6,6)
	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

Q: Probability the sum is 8?
Just count the diagonal!

A: 5 favourable outcomes
→ $\frac{5}{36}$



The Formula:

$$P(E) = \frac{\text{Favourable}}{\text{Total}}$$



The Scale:

$$0 \leq P(E) \leq 1$$

(0 = Impossible, 1 = Sure)



The Opposites:

$$P(E) + P(\text{not } E) = 1$$



Elementary Events:

The sum of all probabilities in an experiment is always exactly 1!



Save this cheat sheet for your exams!